

Sampling People, Networks and Records

Week 3 Quiz: Problem Set

Please input your answers for the following problems via the Coursera course site:

<https://www.coursera.org/learn/sampling-methods>.

From a population of $A = 2,500$ high schools and $N = 500,000$ students an *epsem* sample of students is chosen in $a = 50$ schools with $b = 41$ students selected from each school ($n = 2,050$ students). The *fpc* may be ignored in all calculations.

1. If for a characteristic of interest, $s^2 = 10$, what is the simple random sampling variance of the mean of the characteristic? That is, compute $v(\bar{y}_0) = \frac{s^2}{n}$.
2. If the sampling variance of the same mean is, for the cluster sample, $v(\bar{y}) = 0.01$, then what is the design effect of the mean?
3. What is the rate of homogeneity *roh* for this characteristic? That is, what is $roh = \frac{deff - 1}{b - 1}$?
4. Suppose a new sample is to be selected, and the mean of the same characteristic estimated again from the new sample. If the new sample size is $n = 1,050$, selected as $a = 50$ schools with $b = 21$ students each, then compute a projected simple random sampling variance, using $s^2 = 10$.
5. Compute a projected design effect $deff = 1 + (b - 1)roh$ using *roh* from problem 3 above.
6. Compute a projected sampling variance of the mean for the characteristic from the new sample using $v(\bar{y}) = deff \times v(\bar{y}_0)$.